

1. Classification of Automobile

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* Automobile :-

It is self propelled vehicle driven by an I.C engine & used for transportation of passenger and goods on ground

e.g. Bus, truck.

car

Jeep

* Automobile Engg :-

It is branch of engg in which we study all about the automobile & have practice to propel them

* Classification :-

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① Based on purpose :-

① Passenger vehicles - Bus, car, mopeds

② Goods vehicles - Truck, container, tempo, travellex.

② Based on capacity :-

① Heavy transport vehicles - container, truck for goods

② Light transport vehicles - Tempo, auto

for passenger

③ Light motor vehicle - Car, jeep

④ Medium vehicle - minibus, jeep

③ Based on fuel used :-

① Petrol engine vehicles - small duty 4 wheels & all 2 wheels

② Diesel - " - truck, Buses

ADCET

- (c) Gas vehicles - (d) Solar vehicles -
 (e) Hydrogen - " - (f) Electric - " -
 (g) Hybrid - " - eg Hybrid buses, Hybrid cars

④ Based on construction :-

- (a) Single unit vehicles
 (b) Articulated vehicles & tractors.

⑤ Based on no of wheel :-

- (a) Two wheelers (b) Three wheelers
 (c) four wheelers (4x2 & 4x4)
 (d) Six wheelers (6x2 & 6x4)

⑥ Based on side of drive :-

America - (a) Left hand drive (b) fluid drive
 India & Australia (b) right - " -

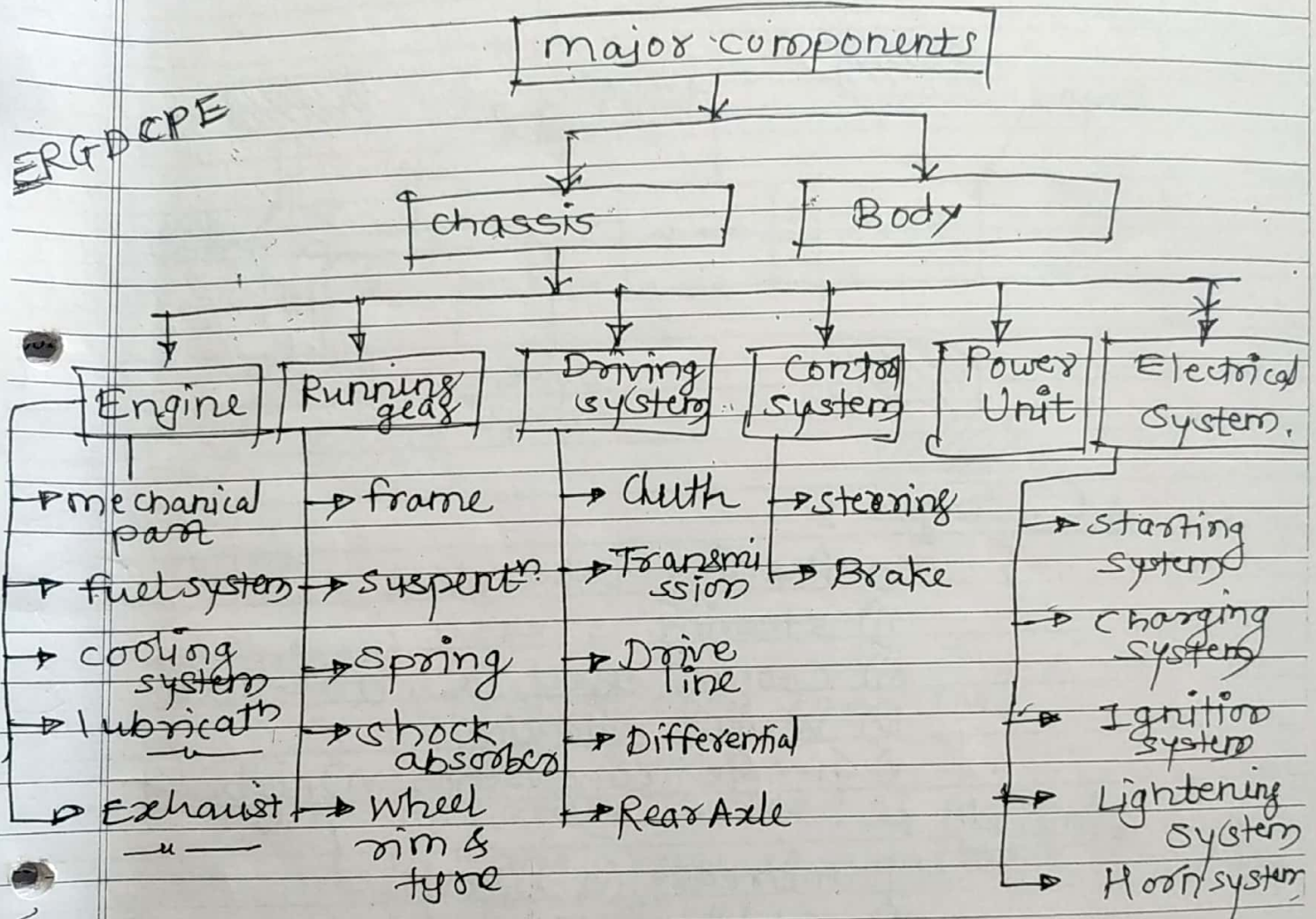
⑦ Suspension system:

- (a) Conventional - Leaf spring
 (b) Independent - Coil spring, torsion bar

⑧ Body & No of doors:

- (a) sedan - two doors, four doors
 (b) Convertible - Jeep, some cars
 (c) Station wagon.
 (d) Special purpose vehicles
 - Ambulance, milkvan.

* Major components of automobile :-



* Vehicle Layout

front engine rear wheel drive. (for syllabus)
Rear ——— front ———

front ——— front ——— (for syllabus)

All wheel drive. (for syllabus)

Heavy vehicle layout.

Articulated vehicle layout

Introduction -

Automobile Engineering is a branch of engineering which deals with everything about automobiles and practices to propel them.

Components of an Automobile -

The main units of an automobile are;

1. The basic structure -

It consists of the frame, the suspension system, axles, wheels and tyres.

a) Frame - There are two distinct forms of construction in common use;

1. The conventional pressed steel frame to which all the mechanical units are attached and on which the body is superimposed.

2. The integral or frameless construction, in which the body structure is so designed as to combine the functions of body and frame. The units normally attached to the frame & then being attached directly to the body.

2. Suspension System -

The objects of suspension are;

1. To prevent the road shocks from being transmitted to the vehicle components.

2. To safeguard the occupants from road shocks.

There are two distinct types of suspension systems;

1. The conventional system in which the road springs are attached to a rigid beam axle.

2. The independent system in which there is no rigid axle beam & each wheel is to free move

Vertically without any reaction on the other wheel.

3. Axles -

The weight carrying portions of the axles, whether it may be front or rear, may be considered as beams supported at the ends, loaded at two intermediate points.

4. Wheels -

Wire spoked wheels have been used mainly on sports cars, primarily on account of their light weight and quickness in changing the wheel.

Light alloy wheels are currently used in case of luxury and sports cars and are called "formula wheels".

* The Power Plant -

The power plant provides the motive power for all the various functions which the vehicle or any part of it.

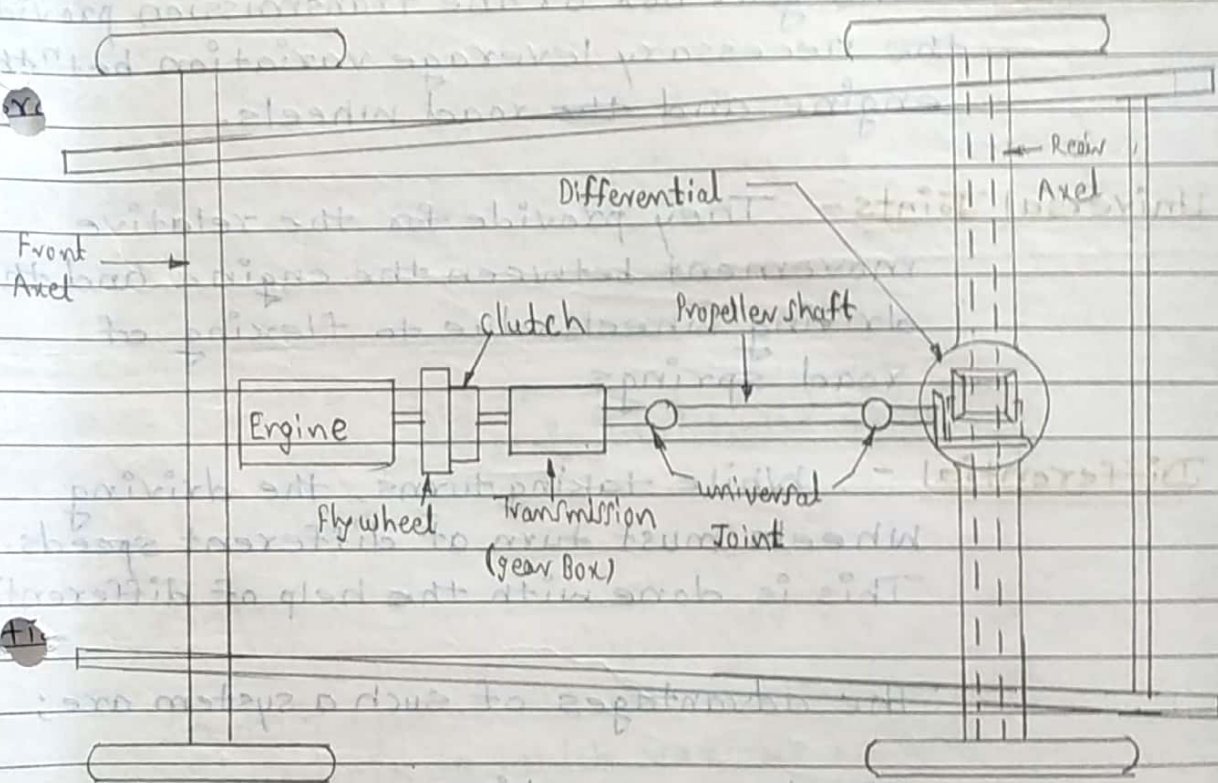
The power plant generally consist of an internal combustion engine which may be either of spark ignition or of compression ignition type.

Electric motors powered by batteries have been used as motive power plant for automobiles. Combination of an I.C. engine and an electric motor have been successfully employed in hybrid vehicles.

Solar energy-powered systems have also been tried, but the most promising future power plant is the one based on fuel cells, which would give zero-emission vehicles.

The Transmission System

The transmission system consist of a clutch, a gear box giving four, five or even six different ratios of torque output to torque input, a propeller shaft to transmit the torque output from the gear box to the rear axle and a differential gear to distribute the final torque equally between the driving wheels.



The functions of a transmission system are;

1. To disconnect the engine from the road wheels when desired.
2. To connect the engine to the driving wheels without shock.
3. To vary the leverage betⁿ the engine & driving wheels.

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The various components which perform these functions are;

- 1) Clutch - Its purpose is to enable the driver to disconnect the drive from the road wheels instantaneously and to engage drive from the engine to the road wheels gradually while moving the vehicle from rest.
- 2) Gear box - The gear box or the transmission provides the necessary leverage variation betⁿ the engine and ~~the~~ road wheels.
- 3) Universal Joints - They provide for the relative movement between the engine and the driving wheels due to flexing of road springs.
- 4) Differential - While taking turns, the driving wheels must turn at different speeds. This is done with the help of differential.

The advantages of such a system are;

1. Simple construction.
2. No torque steer.
3. The front axle is not burdened with steering and power transmitted both. Due to this the suspension of each axle may be optimized for each specialized purpose.

* Frame construction:-

* Materials for frame:-

Carbon - 0.25 - 0.35 %

Mn - 0.35 - 0.75 %

Si - 0.30 %

Ni - 3 %

P - 0.05 %

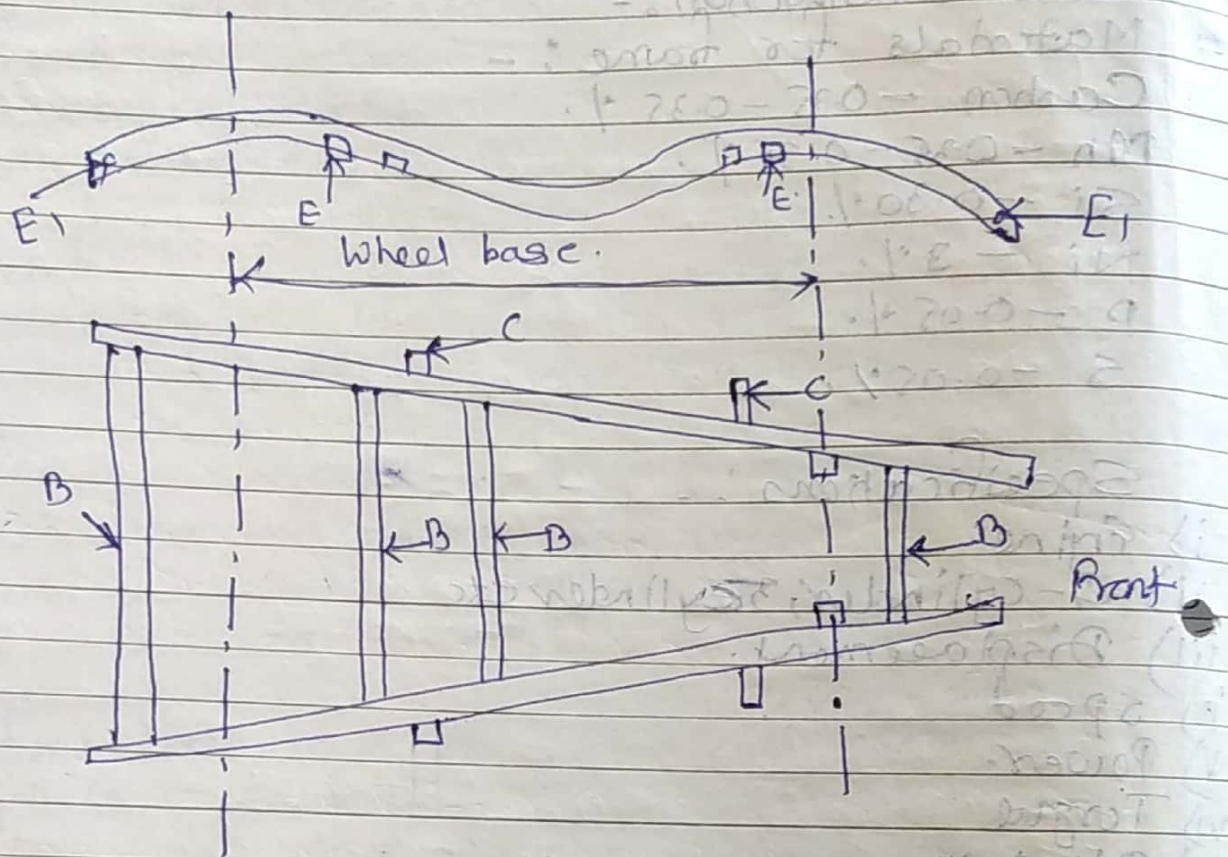
S - 0.05 %

Specifications:-

- i) Engine
- ii) 4-Cylinder, 3-Cylinder etc
- iii) Displacement
- iv) Speed
- v) Power
- vi) Torque
- vii) Drivetrain
- viii) Suspension
- ix) Fuel tank capacity
- x) Wheelbase
- xi) Turning radius
- xii) Dimensions
- xiii) Braces

Frame construction:-

A simplified diagram representing the frame shows the longitudinal members A and the cross members B. The frame is upswept at the rear and front to accommodate the movement of the axles due to springing. It also keeps the chassis height low. The frame is narrowed down at the front as shown in fig. to have a better steering lock, which gives a smaller turning circle. C are the brackets supporting the body. E are the dumb irons to act as bearings for spring shackles. They also take the bumper brackets. Brackets



are meant for mounting the springs. The extension of the chassis frame ahead of the front axle is called front overhang, whereas its extension beyond the rear axle is called rear overhang.

Since the commercial vehicles have to carry large loads, framed construction is invariably used for these. Because in these vehicles, ground clearance is larger and sufficient space is otherwise available for steering the vehicles, the frames for these have only straight members.

The engine, clutch and the transmission are bolted together to form one rigid assembly which is mounted usually on the front end of the frame. It is supported on the frame at three places by means of rubber blocks. This helps to isolate the engine from road shocks and the body from the engine vibrations.

Sub frames:-

Normally various components are bolted directly to the main frame. But many a time, these components are mounted on a separate frame called sub-frame. This sub-frame is further supported by main frame at three points.

Advantages:-

1. The mass of the sub-frame alone helps to damp vibrations.
2. The provisions of sub-frame simplifies production on the assembly line and facilitates subsequent overhaul or repair.

* Resistance to vehicle motion:-

i) Air Resistance:-

This is the resistance offered by air to the movement of a vehicle.

It has influence on performance, ride & stability of vehicle & depends upon the size & shape of the body of the vehicle, its speed & wind velocity.

Generally it is calculated as

$$R_a = K_a A V^2$$

R_a = Air resistance.

A = frontal projected area m^2 .

V = Speed of the vehicle km/hr .

K_a = Coefficient of air resistance.

= 0.023 for best streamline cars

= 0.031 for avg. cars.

= 0.045 for trucks & lorries.

ii) Rolling resistance:-

It depends mainly on

- Nature of road surface.
- Types of tyre.
- wt. of vehicle.

speed of vehicle.

It is expressed as

$$R_r = k_r W$$

W = Weight of vehicle.

k_r = Const. of rolling resistance & depends upon road surface & types of tyre

= 0.05 for good roads

= 0.18 for loose sand roads

= 0.015 Generally used value.

Generally used expression for rolling resistance.

$$R_r = (a + bv) W$$

where V = Speed of the vehicle.

a & b are 0.015 & 0.00016 respectively

iii) Grade resistance:-

The component of the weight of the vehicle parallel to the gradient or slope on which it moves is termed as 'Grade Resistance'. Thus it depends upon the steepness of the grade.

If gradient is expressed in 1 in 5, it means that for every 5 metre the vehicle moves, it is lifted by 1 metre.

It is expressed as

$$R_g = W \sin \alpha$$

W = Total weight of the vehicle.

α = Inclination of the slope to the horizontal.

* Power required for vehicle propulsion.

The motion of a vehicle moving on a road is resisted by aerodynamic forces known as wind or air resistance & road resistance which is generally termed as rolling resistance. In addition to these two types the vehicle has to overcome grade resistance, when it moves

on a gradient, because the weight of the vehicle is to be lifted through a vertical distance. Hence power required to propel a vehicle is proportional to the total resistance to its motion & the speed.

Power required to propel the vehicle is given by eqⁿ

$$P_v = \frac{R \cdot V}{1000} \times \frac{1000}{60 \times 60} \text{ kW}$$

$$= \frac{R \cdot V}{3600} \text{ kW}$$

Where $R = (R_a + R_r)$ when vehicle moves along a level road.

$R = (R_a + R_r + R_g)$ when vehicle moves up a gradient.

Engine power required

$$P_R = \frac{P_v}{\eta_t} = \frac{R \cdot V}{3600 \eta_t} \text{ kW}$$

Where, P_v = Power req. by vehicle.

P_R = engine power required.

V = Speed of vehicle km/hr.

η_t = Transmission or drive line eff.

R = Total resistance.

R_a = Air resistance.

R_r = Rolling resistance.

R_g = Grade resistance.

Traction & Tractive effort:

The force available at the contact between the drive wheel tyres & road is known as tractive effort.

The ability of the drive wheels to transmit this effort without slipping is known as traction.

$$\text{Engine Power} = P_E = \frac{2\pi T N T_E}{6000} \text{ kW}$$

$$\therefore \text{Engine Torque} = T_E = \frac{60000 P_E}{2\pi N} \text{ N.m.}$$

Torque at drive wheels

$$T_w = (g.r \times a.r) \eta_t \cdot T_E$$

$$= G \cdot \eta_t \cdot T_E \cdot \text{N.m.}$$

$$\text{Tractive effort} = F = \frac{T_w}{r} = \frac{T_E \cdot G \cdot \eta_t \cdot N}{r}$$

P_E = Engine Power in K.w.

T_E = Mean engine torque, N.m.

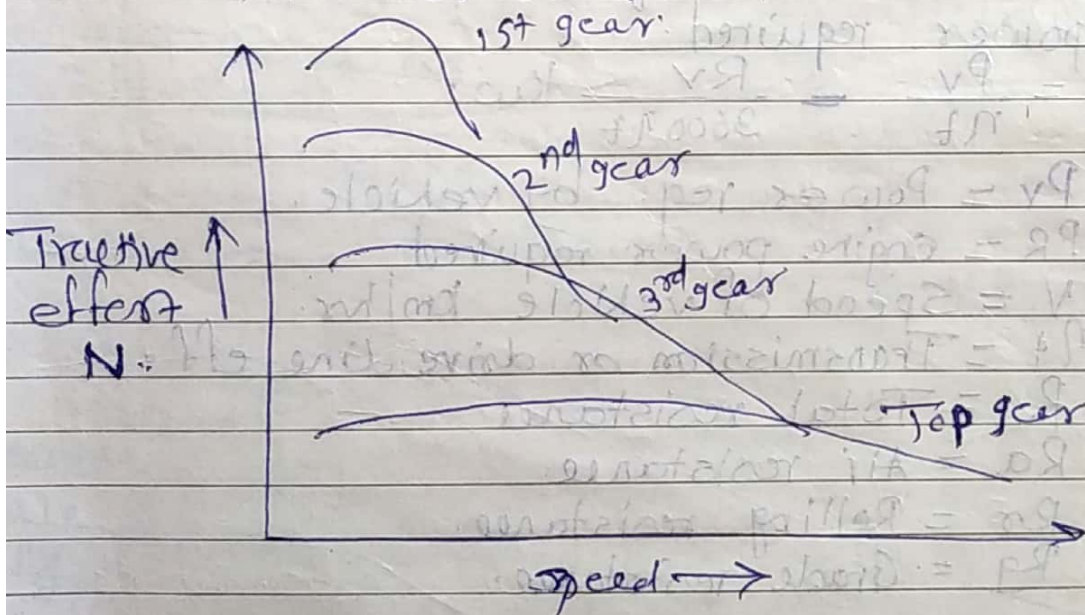
η_t = overall transmission efficiency.

$g.r$ = gear ratio.

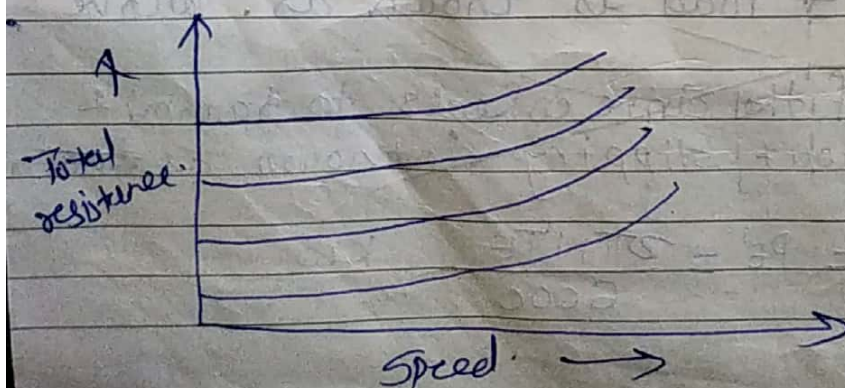
$a.r$ = axle ratio.

G = Overall gear ratio = $(g.r \times a.r)$.

N = R.P.M. of crankshaft.

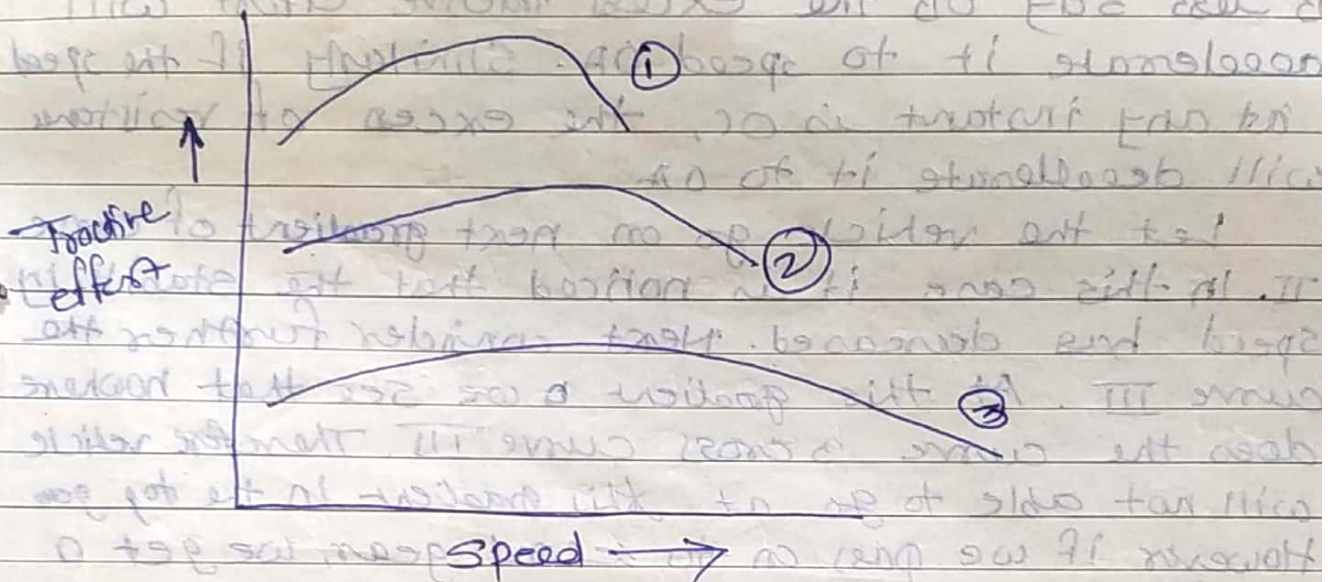


Vehicle Performance curve:-

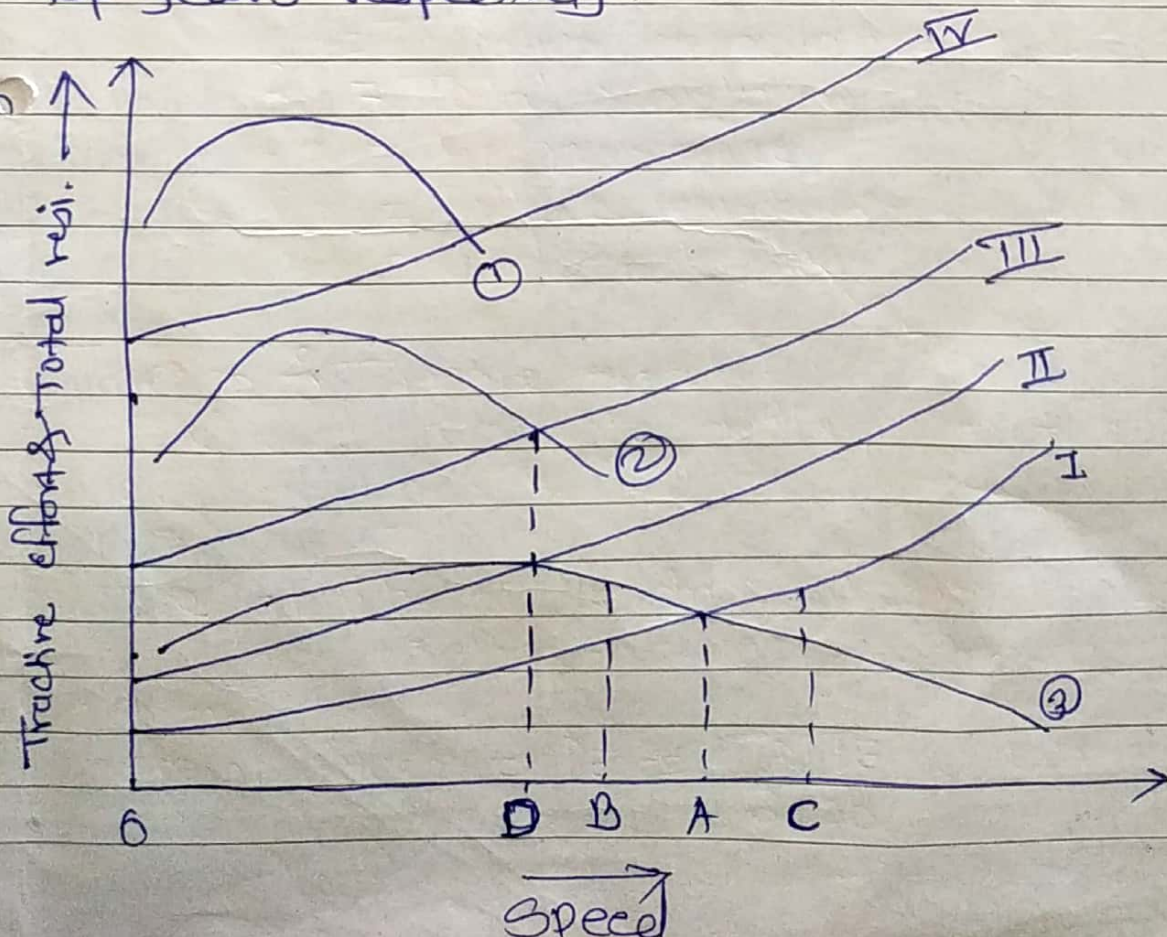


The total resistance for same type of road with different gradients may then be represented by curves in fig. The higher curve represents steeper gradients.

Tractive effort :-



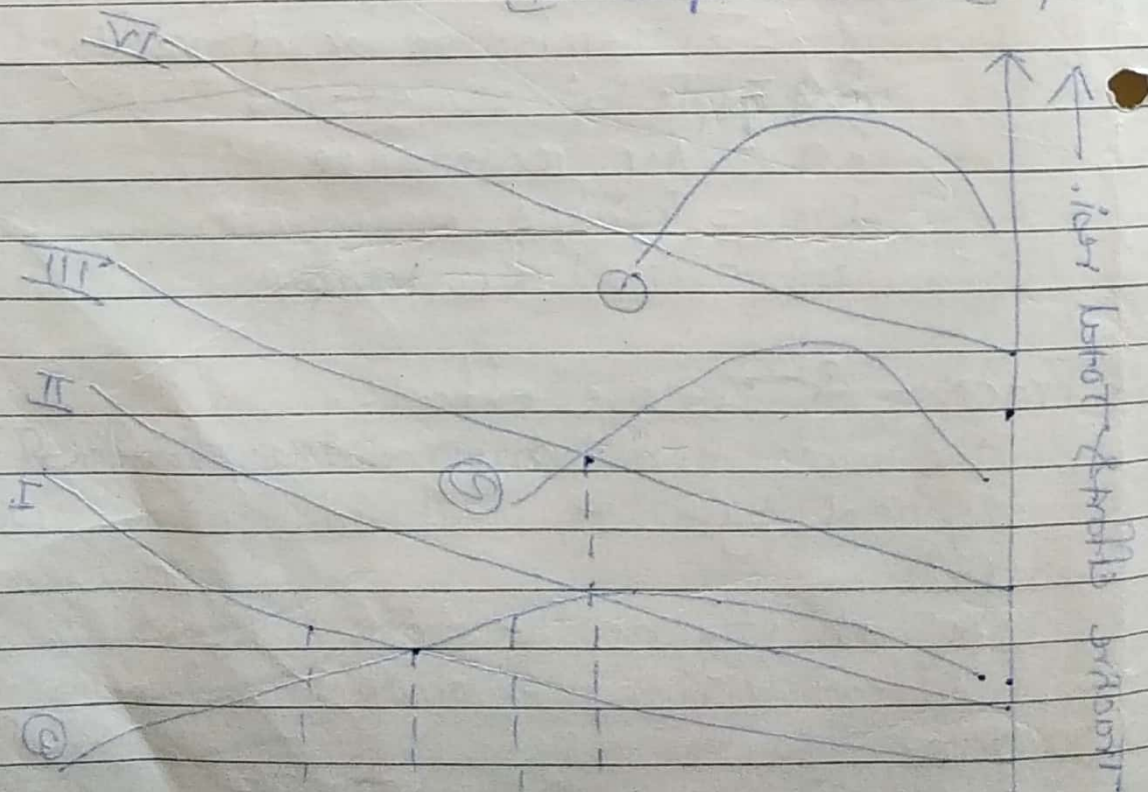
The curves 1, 2 and 3 respectively in above fig. represents the tractive effort in first, second and top gears respectively.



For further clarification consider the curve obtained by combining two curves.

Let the vehicle be in top gear and suppose the vehicle is travelling on a gradient which gives total curve I. Then it is seen that OA is a stabilizing speed. If the speed at any instant is less say OB the excess tractive effort will accelerate it to speed OA. Similarly if the speed at any instant is OC, the excess of resistance will decelerate it to OA.

Let the vehicle go on next gradient of curve II. In this case it is noticed that the stabilizing speed has decreased. Next consider further the curve III. At this gradient we see that nowhere does the curve cross curve III. Therefore vehicle will not be able to go at this gradient in the top gear. However if we pass on to second gear, we get a stabilizing speed OD. Also on second gear vehicle will not be running on gradient IV for which we shall have to shift first gear.



UNIT 1 CLASSIFICATION OF AUTOMOBILES

1.1 Introduction

Automobile engineering is the one of the stream of mechanical engineering. It deals with the various types of automobiles, their mechanism of transmission systems and its applications. Automobiles are the different types of vehicles used for transportation of passengers, goods, etc. Basically all the types of vehicles works on the principle of internal combustion processes or sometimes the engines are called as internal combustion engines. Different types of fuels are burnt inside the cylinder at higher temperature to get the transmission motion in the vehicles. Most of the automobiles are internal combustion engines vehicles only. Therefore, every mechanical and automobile engineer should have the knowledge of automobile engineering its mechanism and its various applications.

1.2 Definition

- ✓ Automobile engineering is a branch of engineering which deals with everything about automobiles and practices to propel them.
- ✓ Automobile is a vehicle driven by an internal combustion engine and it is used for transportation of passengers and goods on the ground. Automobile can also be defined as a vehicle which can move by itself.
Examples: Car, jeep, bus, truck, scooter, etc.

1.3 Classification of Vehicles

Automobiles or vehicles can be classified on different bases as given below :

On the Basis of Load

- (a) Heavy transport vehicle (HTV) or heavy motor vehicle (HMV), e.g. trucks, buses, etc.
- (b) Light transport vehicle (LTV), e.g. pickup, station wagon, etc.
- (c) Light motor vehicle (LMV), e.g. cars, jeeps, etc.

Wheels

- (a) Two wheeler vehicle, for example : Scooter, motorcycle, scooty, etc.
- (b) Three wheeler vehicle, for example : Autorickshaw, three wheeler scooter for handicaps and tempo, etc.
- (c) Four wheeler vehicle, for example : Car, jeep, trucks, buses, etc.
- (d) Six wheeler vehicle, for example : Big trucks with two gear axles each having four wheels.

Fuel Used

- (a) Petrol vehicle, e.g. motorcycle, scooter, cars, etc.
- (b) Diesel vehicle, e.g. trucks, buses, etc.
- (c) Electric vehicle which use battery to drive.
- (d) Steam vehicle, e.g. an engine which uses steam engine. These engines are now obsolete.

(e) Gas vehicle, e.g. LPG and CNG vehicles, where LPG is liquefied petroleum gas and CNG is compressed natural gas.

Body

On the basis of body, the vehicles are classified as :

- (a) Sedan with two doors
- (b) Sedan with four doors
- (c) Hatchback
- (d) Station wagon
- (e) Convertible, e.g. jeep, etc.
- (f) Van
- (g) Pick up
- (h) Special purpose vehicle, e.g. ambulance, milk van, etc.

Transmission

- (a) Conventional vehicles with manual transmission, e.g. car with 5 gears.
- (b) Semi-automatic
- (c) Automatic: In automatic transmission, gears are not required to be changed manually. It is automatically changes as per speed of the automobile.

Position of Engine

Engine in Front:

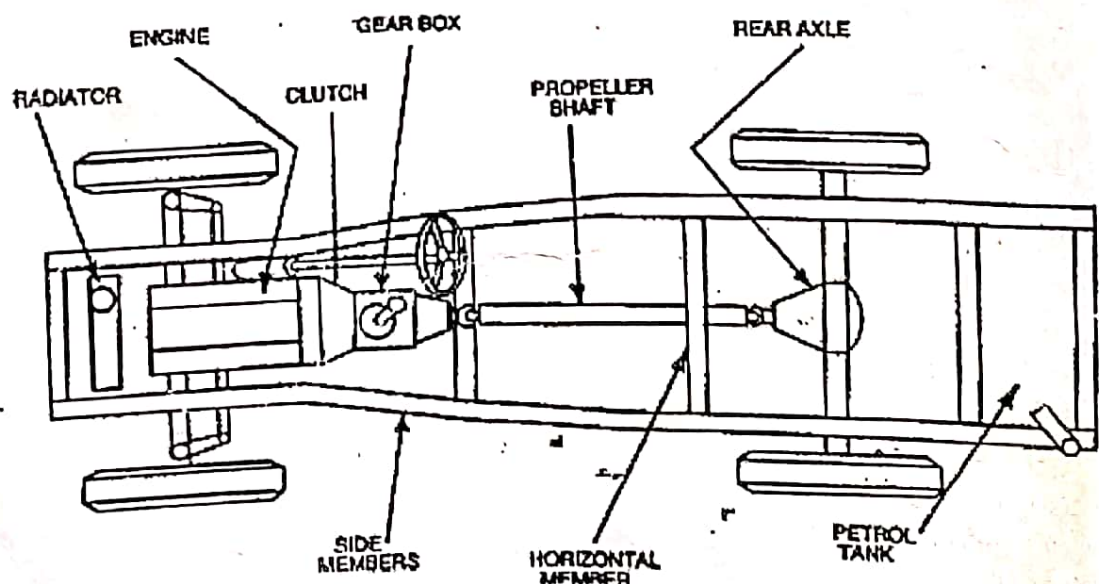
Most of the vehicles have engine in the front. Example : most of the cars, buses, trucks in India.

Engine in the Rear Side:

Very few vehicles have engine located in the rear. Example : Nano car.

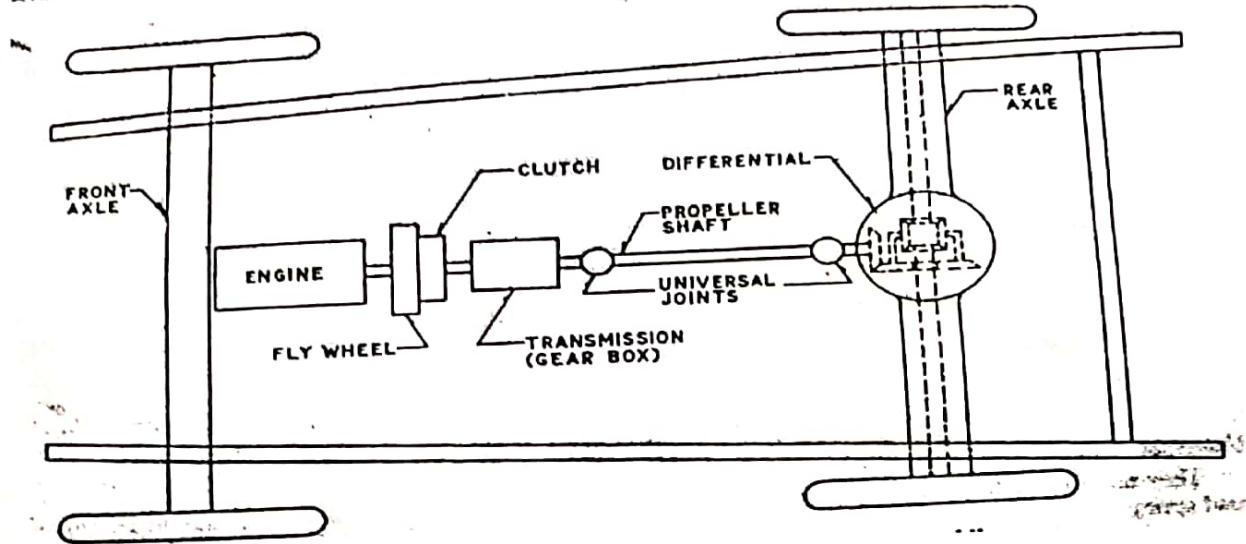
1.4 Layout of An Automobile Chassis

Layout of an engine chassis is shown in the figure given below :



It contains the source of power, i.e. engine, the frame, which supports the engine, wheels, body, transmission, the braking system and the steering. It also gives support to suspension system and springs. Besides these parts

1.5 General Layout of a Transmission system



1.6 Components of the Automobile

The automobile can be considered to consist of five basic components :

- (a) **The Engine or Power Plant:** It is source of power.
- (b) **The Frame and Chassis:** It supports the engine, wheels, body, braking system, steering, etc.
- (c) **The Transmission :** Which transmits power from the engine to the car wheels. It consists of clutch, transmission, shaft, axles and differential.
- (d) **The Body.**
- (e) **The Controls :** It includes steering system, braking system, suspension system which controls the performance of a vehicle
- (e) **Accessories:** It including light, air conditioner/hearer, stereo, wiper, etc.

1.7 Functions of Major Components of An Automobile

Chassis and Frame

The chassis is formed by the frame with the frame side members and cross members. The frame is usually made of box, tubular and channel members that are welded or riveted together. In addition to this, it comprises of the springs with the axles and wheels, the steering system and the brakes, the fuel tank, the exhaust system, the radiator, the battery and other accessories. Along with this the frame supports the body.

Engine or Power Plant

The engine is the power plant of the vehicle. In general, internal combustion engine with petrol or diesel fuel is used to run a vehicle. An engine may be either a two-stroke engine or a four-stroke engine. An engine consists of a cylinder, piston,

valves, valve operating mechanism, carburetor (or MPFI in modern cars), fan, fuel feed pump and oil pump, etc. Besides this, an engine requires ignition system for burning fuel in the engine cylinder.

Transmission System (Clutch and Gear Box)

The power developed by the engine is transferred to the wheels by transmission system. Transmission system must do three jobs:

- (a) It must provide varying gear ratios. Number of gear ratios is equal to number of gears in a vehicle.
- (b) It must provide a reverse gear for moving vehicle in reverse direction.
- (c) It must provide a neutral or disconnecting arrangement so that the engine can be uncoupled from the wheels of the vehicle. In a conventional transmission system, there is a clutch, a manually operated transmission (gear box), a propeller shaft and a differential or final drive.

Clutch

The purpose of the clutch is to allow the driver to couple or decouple the engine and transmission. When clutch is in engaged position, the engine power flows to the transmission through it (clutch). When gears are to be changed while vehicle is running, the clutch permits temporary decoupling of engine and wheels so that gears can be shifted. In a scooter, the clutch is operated by hand where as in a car the clutch is operated by foot. It is necessary to interrupt the flow of power before gears are changed. Without a clutch, it will be very difficult.

Final Drive

Final drive is the last stage in transferring power from engine to wheels. It reduces the speed of the propeller shaft (drive shaft) to that of wheels. It also turns the drive of the propeller shaft by an angle of 90° to drive the wheels.

Braking System

Brakes are used to slow down or stop the vehicle. Hydraulic brakes are generally used in automobiles, where brakes are applied by pressure on a fluid. Mechanical brakes are also used in some vehicles. These brakes are operated by means of levers, linkages, pedals, cams, etc. Hand brake or parking brake is usually a mechanical brake. These are used for parking the vehicles on sloppy surfaces and also in case of emergency.

Gear Box

Gear box contain gearing arrangement to get different speeds. Gears are used to get more than one speed ratios. When both mating gears have same number of teeth, both will rotate at same number speed. But when one gear has less teeth than other, the gear with less number of teeth will rotate faster than larger gear. In a typical car, there may be six gears including one reverse gear. First gear gives low speed but high torque. Higher gears give progressively increasing speeds. Gears are engaged and disengaged by a shift lever.

Steering System

In front wheels can be turned to left and right by steering system so that the vehicle can be steered. The steering wheel is placed in front of driver. It is mechanically linked to the wheels to provide the steering control. The primary function of the steering system is to provide angular motion to front wheels so that vehicle can negotiate a turn. It also provides directional stability to vehicle when the

vehicle moves ahead in straight line. Now-a-days, many vehicles are equipped with power steering which uses pressure of a fluid to reduce steering effort. When driver turns the steering wheel, a hydraulic mechanism comes into play to provide most of the effort needed to turn the wheel.

Front Axle

Front wheels are mounted at the end of front axle. A part of the weight of vehicle is transmitted to the wheels through this axle. The front axle performs several functions. It carries the weight of the front of the vehicle and also takes horizontal and vertical loads when vehicle moves on bumpy roads. When brakes are provided on front wheels, it endures bending stresses and torsional stresses. It is generally made from steel drop forging. It is robust in construction.

Suspension System

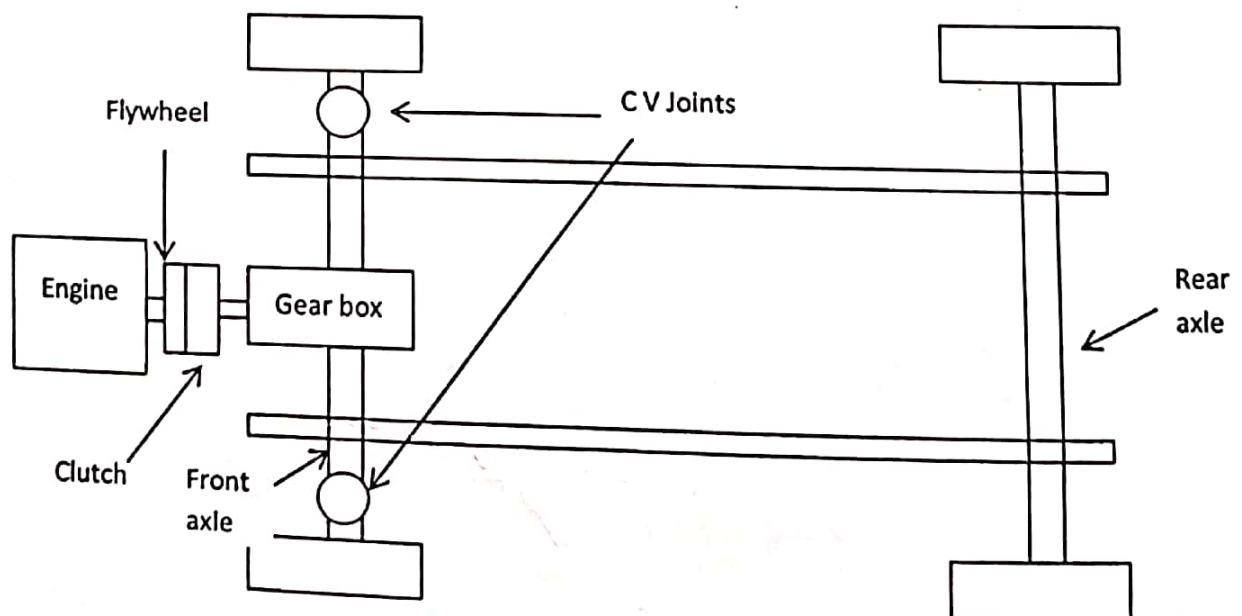
Suspension system of an automobile separates the wheel and axle assembly of the automobile from its body. Main function of the suspension system is to isolate the body of the vehicle from shocks and vibrations generated due to irregularities on the surface of roads. Shock absorbers are provided in the vehicles for this purpose. It is in the form of spring and damper. The suspension system is provided both on front end and rear end of the vehicle. A suspension system also maintains the stability of the vehicle in pitching or rolling when vehicle is in motion.

1.8 Types of Vehicle Layouts

- i) Front engine front wheel drive
- ii) Front engine rear wheel drive
- iii) Rear engine rear wheel drive
- iv) Four wheel drive

Front engine front wheel drive

In this arrangement the engine is located at front of the vehicle & drive is also given to front wheels.



Layout of Front Engine Front Wheel Drive

Advantages

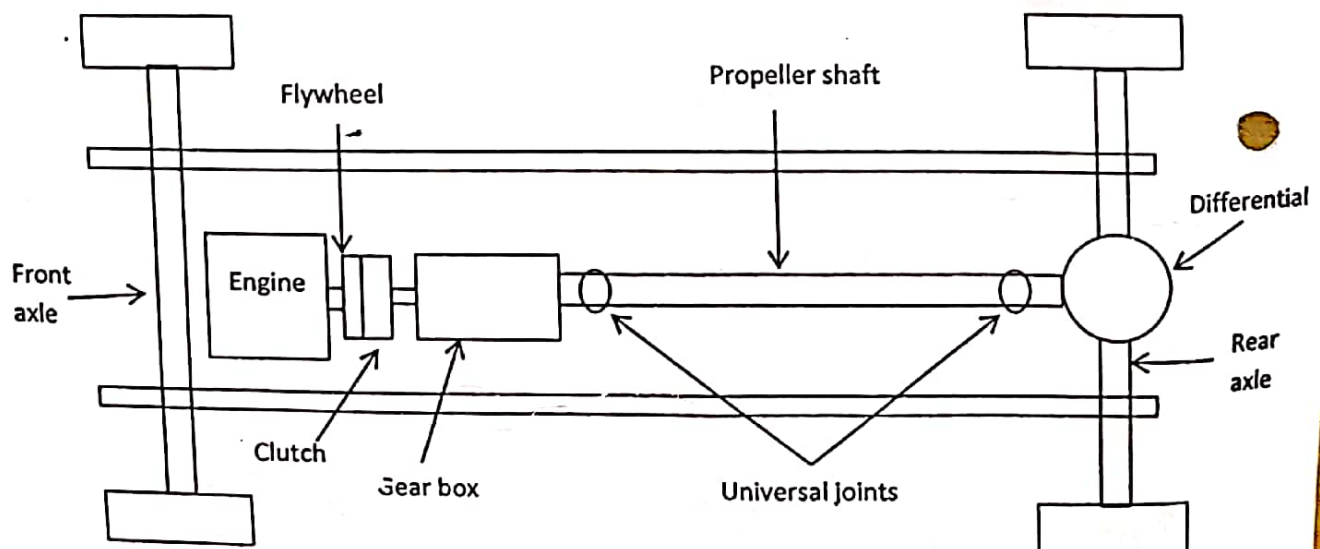
- Propeller shaft, universal joints are eliminated so reduction in cost.
- As propeller shaft is eliminated so low chassis height can be provided which improves the stability of vehicle due to lowered centre of gravity.
- Good road adhesion is obtained due to a large part of the vehicles' weight being carried on the driving wheels under normal conditions
- As weight of engine clutch, gear box and final drive are situated on front axle due to this understeer characteristic is produced which is preferred over the oversteer characteristic.
- Engine pulls the vehicle rather pushing it which avoids skidding tendency & thus increases stability.
- As engine is placed at front side natural cooling improves the cooling efficiency.
- As engine & transmission system components are placed at front so there is increase in boot space.

Disadvantages

- Due to the combination of steered and driven wheels steering system makes complicated.
- Special types of universal joints i. e. constant velocity (C V) joints are required.
- The tractive effort which most needed on steep gradients and during acceleration is reduced.
- As engine is at front so passenger compartment experience some heat & noise.
- Visibility is also hampered somewhat as engine & transmission system components are at front of the driver's cabin.

Front engine rear wheel drive

In this arrangement engine is located at front & drive is transmitted to the rear wheels.



Layout of Front Engine Rear Wheel Drive

Advantages

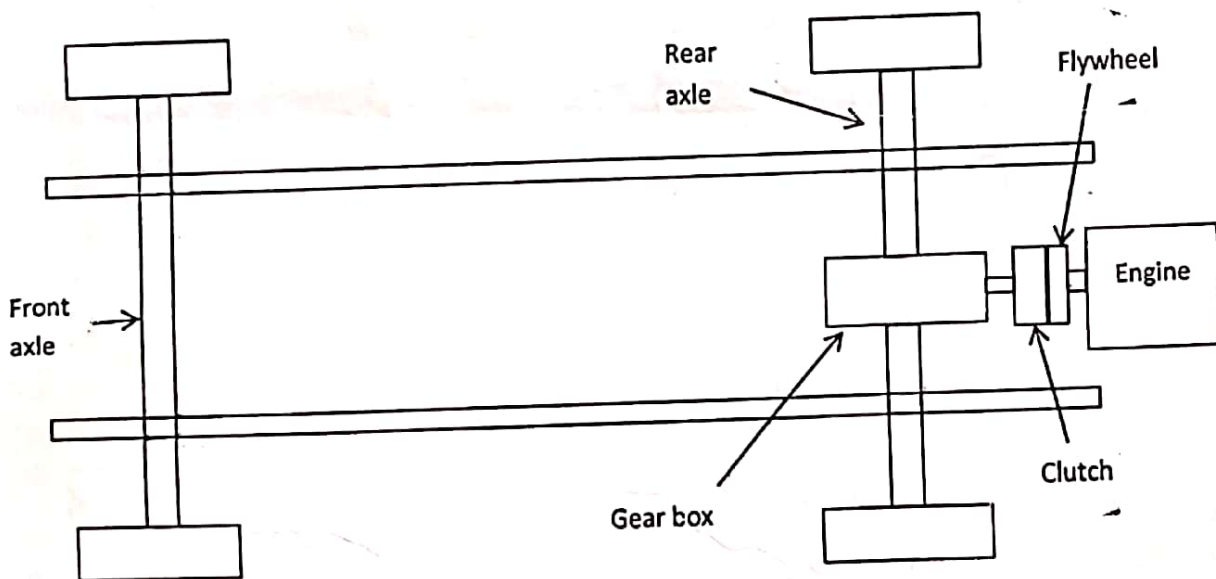
- (a) Reasonably balanced weight distribution between the front and the rear axles providing good handling characteristics
- (b) Front wheel steering arrangement is simplified.
- (c) Behind the rear seats, large luggage space is available providing increased carrying capacity as well as space for easy body movement
- (d) Accessibility to various components like engine, gear box and rear axle is better in comparison to other layouts
- (e) The tractive effort which most needed on steep gradients and during acceleration is increases due to increase in weight on drive wheels.
- (f) As engine is placed at front side natural cooling improves the cooling efficiency.

Disadvantages

- (a) Propeller shaft, universal joints are required to transmit power
- (b) Due to use of the propeller shaft the chassis height is increased which hampers the stability of a vehicle.
- (c) As engine is at front so passenger compartment experience some heat & noise.

Rear Engine Rear Wheel Drive

This arrangement eliminates the necessity for a propeller shaft when the engine is mounted adjacent to the driven wheels. The engine-clutch-gearbox-is mounted at rear side & drive is also given to rear wheels.



Layout of Rear Engine Rear Wheel Drive

Advantages

- (a) It has a simpler steering system arrangement in comparison to front wheel drive.
- (b) The weight of rear engine on the driving wheels provides excellent traction and grip due to increase in tractive effort especially on steep hills as well as when accelerating.
- (c) Effective rear wheel braking is possible with this layout.
- (d) Due to the absence of the propeller shaft the chassis height is reduced which improves the stability.

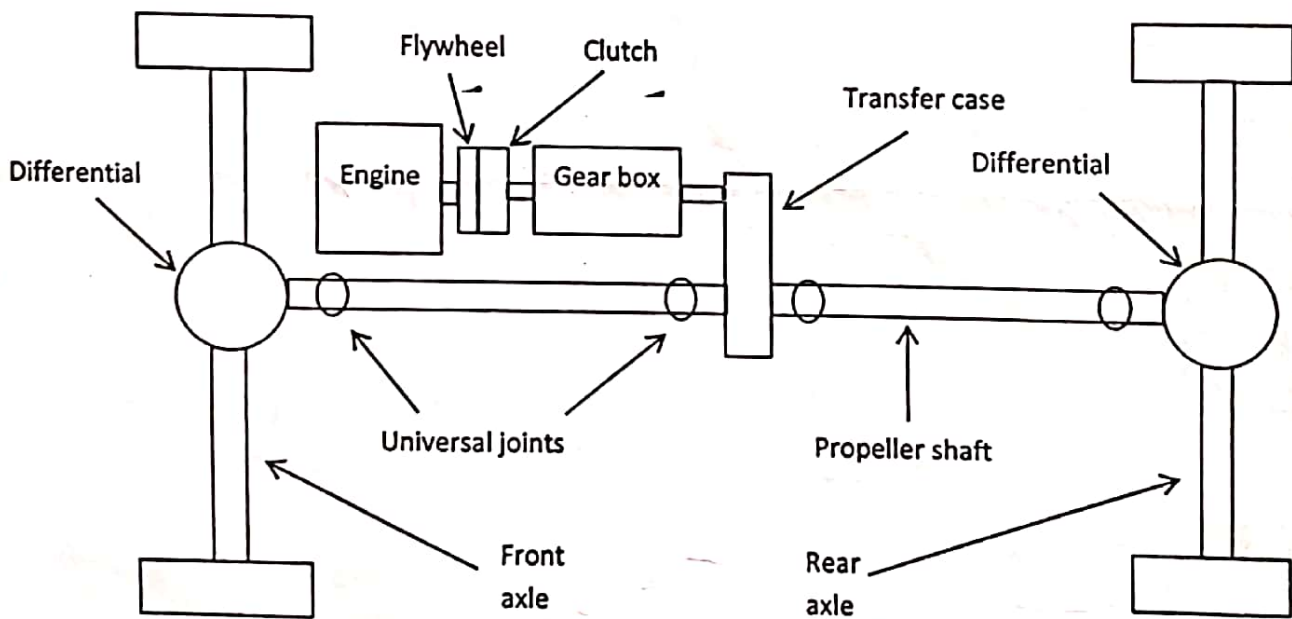
- (e) The front of the vehicle can be designed for good visibility and smooth air flow.
- (f) The exhaust gases, fumes, engine heat and noises are carried away from the passengers.
- (g) This drive arrangement results in compact layout and short car.

Disadvantages

- (a) It has a restricted luggage compartment due to narrow front, which houses the fuel tank also.
- (b) Natural air cooling is not possible, it requires a powerful fan.
- (c) Long linkages are required for the engine, clutch and gear box controls.
- (d) The rearward concentration of weight causes the vehicle to be more affected by side winds at high speeds. This makes the vehicle unstable resulting in oversteer characteristics.

Four-Wheels Drive

To increase manoeuvrability of the vehicle required to travel on rough unconstructed roads and trucks another arrangement known as Four-wheel drive is provided. Due to all the four wheels getting driven, whole weight of the vehicle is available for traction. But this advantage is not worth the additional cost on good road surfaces. The system is provided in jeeps which are known as 4 x 4 wheel drive vehicles.



Layout of Four-Wheels Drive

Any wheel of a vehicle can come out from a ditch or from slippery surface as all wheels are power driven so it is mainly used in military vehicles and sports cars. Some drawbacks are increased cost as use of special device such as transfer case & reduced fuel economy as all wheels are supplied with drive.

1.9 Functions of The Chassis Frame:

- ✓ 1. To carry load of the passengers or goods carried in the body.
- ✓ 2. To support the load of the body, engine, gear box etc.,
- ✓ 3. To withstand the forces caused due to the sudden braking or acceleration
- ✓ 4. To withstand the stresses caused due to the bad road condition.
- ✓ 5. To withstand centrifugal force while cornering

1.10 Types of Chassis Frames:

There are three types of frames

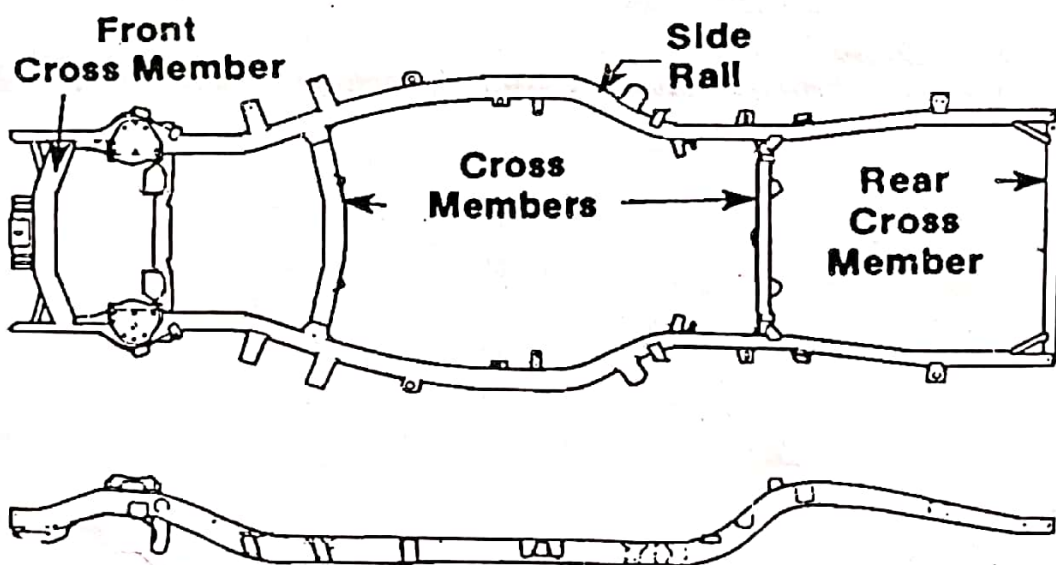
1. Conventional frame
2. Integral frame
3. Semi-integral frame

1. Conventional frame:

✓ It has two long side members and 5 to 6 cross members joined together with the help of rivets and bolts.

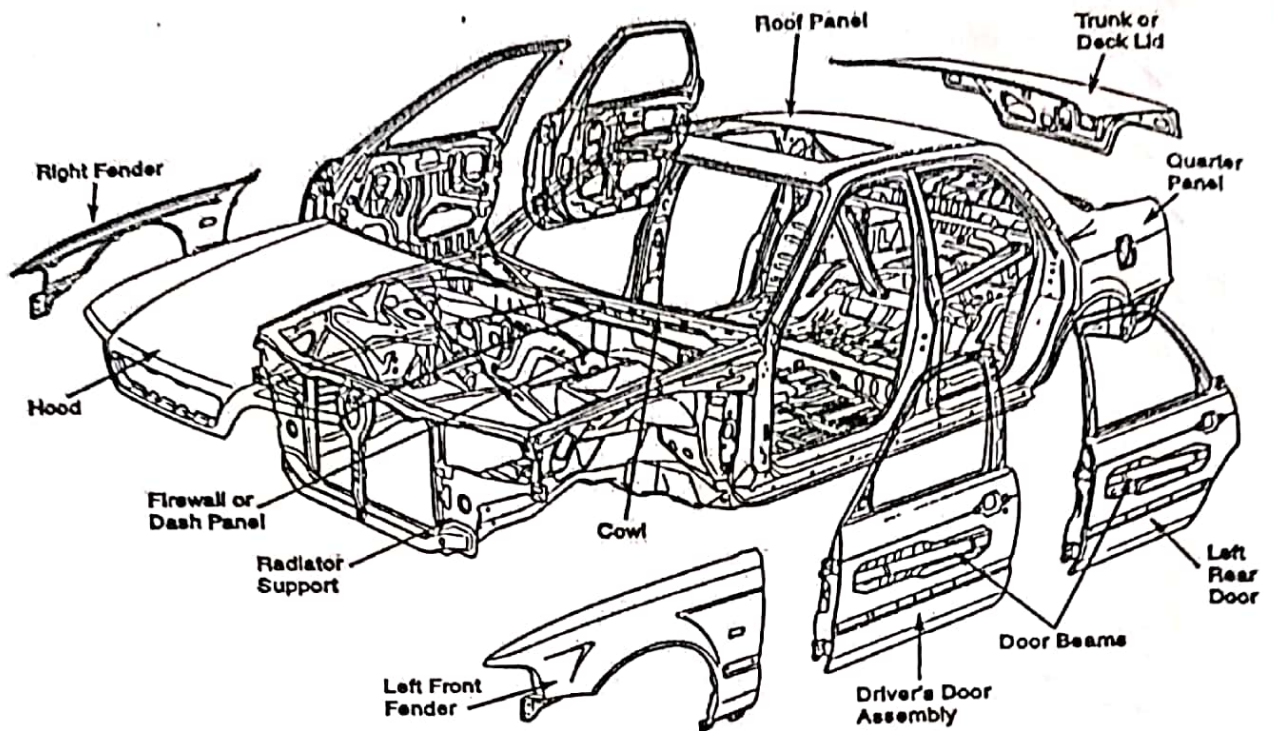
The frame sections are used generally.

- ✓ a. Channel Section - Good resistance to bending
- ✓ b. Tabular Section - Good resistance to Torsion
- ✓ c. Box Section - Good resistance to both bending and torsion



2. Integral Frame:

This frame is used now a days in most of the cars. There is no frame and all the assembly units are attached to the body. All the functions of the frame carried out by the body itself. Due to elimination of long frame it is cheaper and due to less weight most economical also. Only disadvantage is repairing is difficult.



3. Semi - Integral Frame:

In some vehicles half frame is fixed in the front end on which engine gear box and front suspension is mounted. It has the advantage when the vehicle is met with accident the front frame can be taken easily to replace the damaged chassis frame. This type of frame is used in FIAT cars and some of the European and American cars.

1.11 Various Loads Acting On The Frame:

Various loads acting on the frame are

1. Short duration Load - While crossing a broken patch.
2. Momentary duration Load - While taking a curve.
3. Impact Loads - Due to the collision of the vehicle.
4. Inertia Load - While applying brakes.
5. Static Loads - Loads due to chassis parts.
6. Over Loads - Beyond Design capacity.

1.12 Different Bodies Used In Automobiles

According to Chassis design the body can divided into

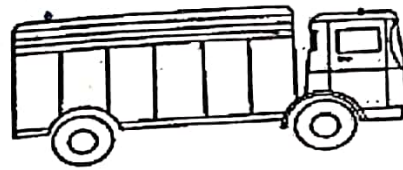
1. Conventional Type
2. Integral Type
3. Semi- Integral Type

According to other usage:

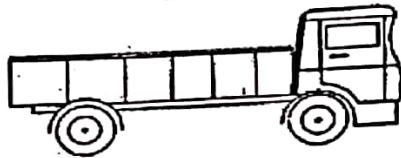
1. Light vehicle Bodies - cars, jeeps
2. Heavy vehicle Bodies - Busses, Lorries
3. Medium vehicle Bodies - Vans, Matador.



1. Car



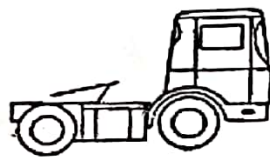
2. Truck Punjab body or Straight truck



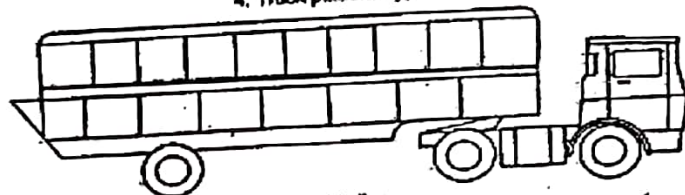
3. Truck half body



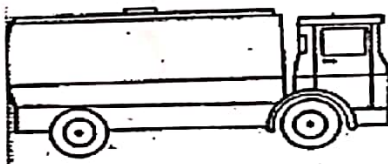
4. Truck platform type



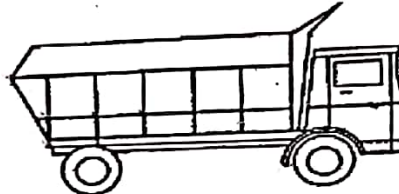
5. Tractor



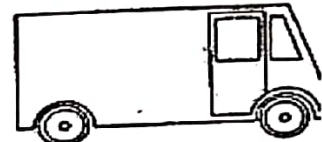
6. Tractor with articulated trailer.



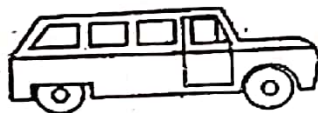
7. Tanker



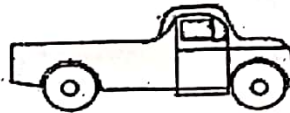
8. Dumper truck



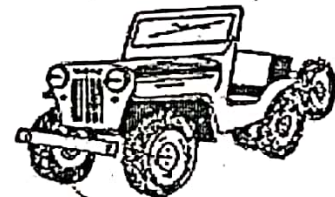
9. Delivery Van



10. Station wagon



11. Pick-up



12. Jeep

1.13 Requirements Of Bodies For Various Types Of Vehicle:

The body of the most vehicles should fulfil the following requirements:

1. The body should be light.
2. It should have minimum number of components.
3. It should provide sufficient space for passengers and luggage.
4. It should withstand vibrations while in motion.
5. It should offer minimum resistance to air.
6. It should be cheap and easy in manufacturing.
7. It should be attractive in shape and colour.
8. It should have uniformly distributed load.
9. It should have long fatigue life
10. It should provide good vision and ventilation.